RAZIK TELECOM ALGIERS, ALGERIA

Commercial & Technical Project Proposal

Turnkey Telecom Credit Distribution ERP & Multi-Channel POS Platform

Scope Reference: Grounded in `docs/Razik_Telecom_Detailed_ERP_Scoping_Report_v0.1.pdf` and the operational baseline delivered by SaNDS Lab Middle East W.L.L.

DOCUMENT METADATA & PROJECT BASELINE	
Document Code	**PROP-RAZIK-2026-V1**
Date of Proposal	October 28, 2026
Prepared For	**Razik Telecom** (Owner & Management)
Prepared By	**SaNDS Lab Middle East W.L.L** (Engineering & Architecture)
Geographic Scope	Algiers & Surrounding Wilayas, Algeria
Telecom Operators	Mobilis (ATM), Djezzy (OTA), Ooredoo (WTA)
Target Solution Channels	Android Mobile APK, Widescreen 1920px Web Portal, Windows POS Terminal
Classification	COMMERCIAL-IN-CONFIDENCE

Executive Proposition: This proposal presents the complete turnkey plan to transform Razik Telecom from an owner-dependent manual recharge operation into an automated, auditable, high-concurrency distribution switch. The system provides retail agents with sub-3-second automated recharges, ensures 100% financial control over post-sale credit and instalment collections, and establishes an extensible foundation for bill utilities and B2B electronics commerce.

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1. Executive Summary & Strategic Context

1.1 Market Context & Client Profile

Razik Telecom is an established B2B telecom credit distributor operating in Algiers, Algeria. The company purchases bulk wholesale mobile airtime and data packages from the three national licensed operators—**Mobilis** (06), **Djezzy** (07), and **Ooredoo** (05)—and distributes them to a broad network of approved retail agents including phone kiosks, cybercafés, neighborhood mini-markets, and multi-service shops.

The current operating model relies on the owner directly handling transaction approvals, credit provisioning, and physical cash collection under post-sale instalment agreements. While this has fostered close client relationships, it imposes significant operational ceilings on volume, speed, and financial control.

1.2 Current Operational Ceilings & Pain Points

ID	Current Operational Bottleneck	Business Risk & Operational Impact	Severity
PP-01	Manual Owner Intervention	Transaction delays of 2–5 minutes; inability to handle peak evening recharge surges; limits total daily revenue.	CRITICAL
PP-02	Manual Operator Selection	Retailers or dispatcher misidentify operator prefixes, leading to misrouted airtime and lost funds.	HIGH
PP-03	Informal Post-Sale Credit & Ageing	Lack of automated credit caps allows agent receivables to exceed recoverable limits without automated warning.	CRITICAL
PP-04	Physical Cash Collection Discrepancies	Collections performed on the road lack instantaneous numbered digital receipts, creating reconciliation friction.	HIGH
PP-05	Operator Balance Visibility Gaps	Lack of real-time low-balance alerts results in sudden stockouts and failed recharge attempts.	HIGH

1.3 Strategic Solution Vision

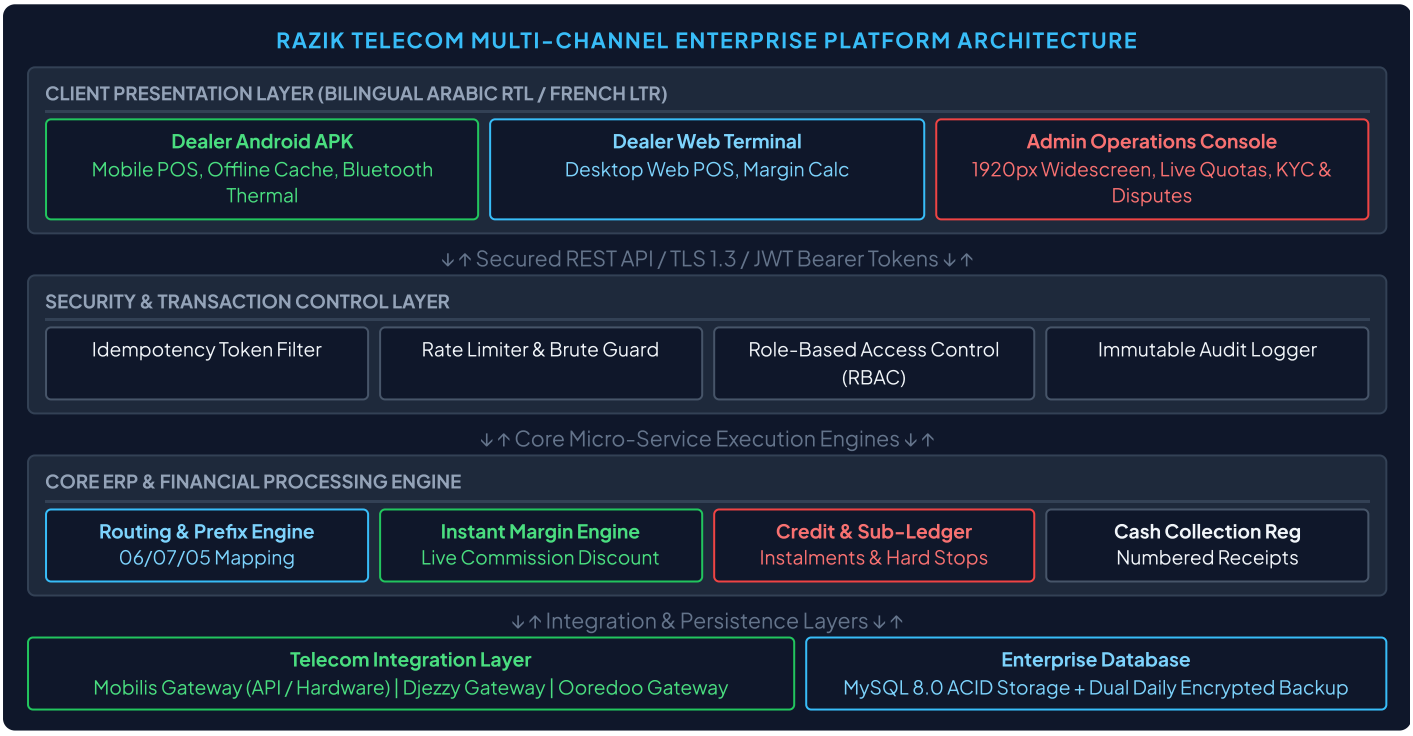
SaNDS Lab Middle East W.L.L has architected the *Razik Telecom Platform* to transition operations from manual dispatching to an **automated, auditable, high-concurrency transaction switch**:

- **Instant Sub-3-Second Recharges:** Retailers submit recharge requests via Android APK or Web POS; the engine auto-identifies the operator via prefix, validates dealer limits, deducts the wholesale net, and executes the recharge automatically.
- **Double-Entry Ledger & Credit Control:** Configurable hard/soft credit limits, automated ageing buckets, and instalment plans with automatic temporary suspension of delinquent retailers.
- **Numbered Digital Cash Collection Receipts:** Field cash collectors issue digitally signed, sequential receipts allocated directly against the oldest pending receivables.
- **Widescreen 1920px Operations Center:** Comprehensive real-time monitoring of wholesale balances, dealer approvals, live margins, and one-click dispute resolution.

2. Strategic Objectives & Success Metrics

Metric / Domain	Current Manual Baseline	Target With Razik ERP	Quantifiable Business Impact
Transaction Turnaround	2 – 5 minutes (manual delay)	**< 3 seconds** (automated)	**50× throughput capacity;** zero manual intervention for standard recharges.
Prefix Accuracy	Human selection error (~3%)	**100% automated mapping**	Zero misrouted transactions across 06 (Mobilis), 07 (Djezzy), 05 (Ooredoo).
Bad Debt Exposure	Informal balance accumulation	**Automated Credit Hard Stop**	System locks sales when overdue thresholds or credit limits are breached.
Cash Collection Audit	Manual paper / WhatsApp notes	**Sequential Digital Receipts**	100% verifiable collection audit trail with collector attribution.
Service Availability	Periodic stockout surprises	**Live Multi-Threshold Alerts**	Automatic low-balance warning (20%) and emergency top-up notifications.

3. System Architecture & Topology



4. Functional Scope Breakdown (The 6 Pillars)

Pillar 1: Platform Foundation, Identity & Access Control

- **Self-Service Dealer Registration:** Prospective retailers submit business credentials, commercial registry number (Registre du Commerce), phone number, and store Wilaya location.
- **Owner Approval Gate:** Submitted accounts remain locked in `PENDING` until the owner verifies identity, assigns initial credit facility, sets commission overrides, and approves activation.
- **Role-Based Access Control (RBAC):** Strict segregation of duties across Super Admin, Kiosk Dealer, Cash Collector, and Support Operator.
- **Enterprise Security:** Session timeout configuration, brute force throttling, password encryption (bcrypt), and complete IP/device audit trail.

Pillar 2: High-Speed Telecom Recharge Engine

- **Automatic Prefix Detection:** Analyzes Algerian phone inputs in real-time: `06` → **Mobilis** | `07` → **Djezzy** | `05` → **Ooredoo**.
- **Instant Commission Discount Calculation:**

> **Live Example:** When a kiosk initiates a **100 DZD** Mobilis recharge with an approved **5.00%** dealer commission rate:
> • End-Customer Receives: **100 DZD** airtime credit | • Dealer Wallet Debited: **95 DZD** net | • Dealer Retains: **5 DZD** cash profit immediately.

- **Idempotency & Duplicate Protection:** Unique UUID tokens per transaction ensure that double-clicking or network drops never trigger multiple debits or duplicate recharges.
- **Controlled Pending Transaction Queue:** Transactions with delayed operator acknowledgments are placed in an active monitoring queue with auto-retry and manual admin resolution controls.

Pillar 3: Receivables, Instalments & Physical Cash Collection Sub-Ledger

- **Double-Entry Ledger:** Every transaction, commission, wallet replenishment, and refund generates an immutable double-entry journal entry with balance snapshots.
- **Instalment Scheduling & Ageing Analysis:** Configurable payment plans with due date tracking; automatic categorization into Current, 30-Day, 60-Day, and 90-Day overdue buckets.
- **Credit Hard/Soft Stops:** Automatic system warnings and conditional transaction locks when an agent exceeds their maximum credit limit or has overdue instalments.
- **Tamper-Proof Cash Collection Register:** Field collectors record cash received on-site; system generates a sequential numbered receipt with automatic allocation against the oldest unpaid invoices.

Pillar 4: Operator Wholesale Quota Monitoring & Reconciliation

- **Real-Time Quota Gauges:** Live visual display of Razik Telecom's remaining wholesale balances with Mobilis, Djezzy, and Ooredoo with Warning Level (20%) and Critical Lockout (5%) thresholds.
- **Daily Reconciliation:** End-of-day discrepancy reporting matching operator consumption against internal ledger records.

Pillar 5: Bilingual User Experience & Multi-Channel Delivery

- **Native Arabic (RTL) & French/English (LTR):** Contextual localized terminology for Algerian telecom terms (Flexy, Arsselli, Storm, Solde, Versement).
- **1920px Widescreen Desktop Operations Center:** High-density operations dashboard featuring Algerian national emblem and live Algiers digital clock (`Africa/Algiers`).
- **Android Mobile APK:** Fast, lightweight Android application designed for kiosk counter tablets and smartphones, with support for Bluetooth thermal receipt printers.

Pillar 6: Future Foundation (Bill Utilities & E-Commerce Ready)

- **Utility Payment Connectors:** Data model and workflow reservations for **Sonelgaz** (Electricity and Gas bill barcode settlement) and **ADE** (Algérienne Des Eaux water bill processing).
- **Physical Accessories E-Commerce:** Pre-built database hooks for phone accessories (chargers, earphones, screen protectors) including stock tracking and delivery routing.

5. Delivered Working Baseline (Proof of Capability)

Unlike traditional software proposals that present abstract concepts, **SaNDS Lab Middle East W.L.L has already engineered and demonstrated a live, functional baseline** directly within the codebase:

Delivered Component	Implemented Engineering Assets	Operational Status	Verification
Relational Database	MySQL 8.0 schema with 9 production tables (`users`, `operators`, `recharges`, `wallet_transactions`, `complaints`, etc.).	ACTIVE & SEEDED	Verified live
REST API Backend	PHP 8.2 API with 8 production endpoints (Stats, Recharge Execution, Dealer Approval, Wallet Adjustments, Refund Resolution).	ACTIVE & TESTED	Port 8000
Admin Operations Web Portal	React 18 + Vite Widescreen (1920px) dashboard with Algerian national theme, live Algiers clock, and multi-operator quota progress bars.	FUNCTIONAL	Port 5173
Dealer POS Simulator	Integrated smartphone kiosk simulator with instant commission calculator, auto-prefix detection, and digital receipt generator.	VERIFIED	Validated live
Mobile App Codebase	React Native / Expo mobile codebase scaffolded in `mobile-app/` with screen flows and API integration layer.	SCAFFOLD READY	EAS Build Ready

6. Phased Implementation Roadmap

Phase	Focus Domain	Key Engineering Deliverables	Timeline	Exit Gate
Phase 0 & 1	Discovery & Prototype	Current-state workflow mapping, interactive prototype, initial schema, and live POS simulator.	COMPLETED	Baseline Approved
Phase 2	Gateway Specification	Telecom modem gateway / API connectivity specification, high-concurrency database locking, security baseline.	Weeks 1 – 2	Architecture Lock
Phase 3	Core ERP & Android Build	Full instalment sub-ledger, cash collection register, native Android APK compilation, thermal printer integration.	Weeks 3 – 7	System UAT Passed
Phase 4	Algiers Kiosk Pilot	Deployment to 10–15 retail kiosks in Algiers, parallel manual running, stress testing, reconciliation validation.	Weeks 8 – 10	Pilot Sign-off
Phase 5	Production Launch	Full production rollout, owner and staff administration training, user guides (Arabic/French), 90-day warranty start.	Weeks 11 – 12	Production Live
Phase 6	Commerce Expansion	Sonelgaz / ADE utility bill integration and B2B physical accessories inventory catalogue.	Post-Launch	Add-on Milestone

7. Technical Specifications & Technology Stack

Technology Tier	Selected Platform	Engineering Rationale
Backend API Framework	PHP 8.2 REST Architecture	Ultra-lightweight, native support across local office servers in Algeria, high execution speed, minimal runtime overhead.
Database Engine	MySQL 8.0 Enterprise Relational	ACID transactional integrity, row-level locking during high-frequency wallet debits, robust double-entry sub-ledger schema.
Operations Web Portal	React 18 + Vite	High responsiveness, native widescreen 1920px multi-monitor support, zero external UI bloat, instant dark/light & RTL switching.
Dealer Mobile POS	React Native / Expo	Cross-platform native compilation to Android APK; supports Bluetooth thermal receipt printers and offline status caching.
Data Security & Privacy	TLS 1.3 / AES-256	Masked customer phone storage in audit logs, salted bcrypt password security, and operator secret encapsulation.

8. Controls, Risk Management & Mitigations

ID	Identified Risk	Impact	Engineered Architectural Mitigation
R-01	Timeout During Recharge	HIGH	**Idempotency Token & Managed Pending Queue:** Recharges are tagged with unique UUIDs. Timeout transactions enter an active pending queue that polls operator status before releasing funds or reversing debits.
R-02	Excessive Retailer Debt	HIGH	**Automated Credit Limits & Ageing Freeze:** The system automatically blocks recharge access if an agent's balance breaches their approved credit ceiling or if instalments are overdue past agreed grace periods.
R-03	Cash Collection Discrepancy	MEDIUM	**Sequential Unalterable Receipts:** Every cash collection creates an immutable, sequentially numbered receipt sent to the dealer; errors can only be resolved via counter-signed contra entries.
R-04	Local Server Failure / Outage	HIGH	**Automated Daily Dual-Backup:** Automated cron tasks perform nightly encrypted database snapshots stored locally and synced to secure off-site cloud storage.
R-05	Ported Number Prefix Misdirection	MEDIUM	**Manual Operator Override Toggle:** While prefix auto-detection is active by default, the dealer POS interface includes a 1-click operator override if a customer has ported their line.

9. Commercial Investment & Milestone Structure

Milestone	Delivered Work Package & Milestone Criteria	Target Timeline	Payment Share
M1	**Contract Execution & Project Kickoff:** Architecture sign-off, scoping lock, environment setup.	Week 1	**20%**
M2	**Core ERP & Financial Engine:** REST API completion, sub-ledger, cash collection register & web portal.	Week 4	**30%**
M3	**Mobile POS & Telecom Integration:** Android APK packaging, Bluetooth thermal printing, and routing connectors.	Week 8	**25%**
M4	**Algiers Field Pilot Sign-off:** Successful 15-kiosk live pilot with verified reconciliations.	Week 10	**15%**
M5	**Production Launch & Handover:** Live production deployment, staff training, operational guides, warranty start.	Week 12	**10%**
Total Turnkey Project Delivery			**100%**

10. Formal Acceptance & Authorization

By signing below, the parties confirm their agreement to the technical scope, delivery roadmap, and commercial terms set forth in this document.

Authorized for Razik Telecom

Name: Razik Telecom Management
Title: Owner & Managing Director

Signature & Official Stamp
Date: _____ / _____ / 2026

Authorized for SaNDS Lab Middle East W.L.L

Name: Lead Enterprise Architect
Title: Engineering Delivery Director

Signature & Official Stamp
Date: _____ / _____ / 2026